RICE BUSINESS · SPRING 2027

Agentic AI: Personal and Enterprise

led by **Kerry Back**

This course is about working with agentic AI on your own tasks and putting it to work inside an organization. It opens with what a large language model actually does, how the current models differ, and how to get dependable results from them on data and documents. It then turns to the machinery underneath — the tools and harnesses an agent runs on, what a sandbox does and does not contain, and the instructions that persist from one session to the next — and to customizing that machinery with skills and plugins you write and distribute. The later weeks build and evaluate AI-enabled software, ground it in your own documents through retrieval and fine-tuning, and end with the work that has to stay in-house: running models on your own hardware and de-identifying data before it goes anywhere else.

WEEK 1	How LLMs work · the universe of models · data analysis and docs · effective use of AI
WEEK 2	Agent tools and harnesses · sandboxed agents · system prompts · persistent instructions
WEEK 3	Creating custom skills · distributing custom skills and scripts as plugins
WEEK 4	Creating AI-enabled software · evaluating models
WEEK 5	Retrieval-augmented generation · fine-tuning small models
WEEK 6	Running models locally · de-identifying tables and text

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